

Nuclear Medicine Notes

Radioactive Activity and Decay Processes

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1 Radioactive Activity

Radioactive activity describes how quickly radioactive nuclei are undergoing nuclear transformations.

1.1 Activity Formula

Activity is defined as

$$A = -\frac{dN}{dt}$$

where

- A = radioactive activity
- N = number of radioactive nuclei
- t = time

The negative sign is present because the number of radioactive nuclei decreases as radioactive decay occurs.

Activity can also be expressed as

$$A = \lambda N$$

where

λ = decay constant.

Therefore,

$$A = -\frac{dN}{dt} = \lambda N$$

2 Decay Constant

The decay constant is represented by the Greek letter

λ .

In radioactive decay, λ represents the **decay constant**.

It should not be confused with the use of λ for wavelength in wave physics.

The decay constant is related to physical half-life by

$$\lambda = \frac{\ln 2}{T_{1/2}}$$

Since

$$\ln 2 \approx 0.693,$$

this can also be written as

$$\lambda = \frac{0.693}{T_{1/2}}$$

2.1 Units of the Decay Constant

The units of λ depend on the time unit used.

If time is measured in seconds,

$$\lambda \text{ has units of } \text{s}^{-1}.$$

If time is measured in hours,

$$\lambda \text{ has units of } \text{h}^{-1}.$$

For example,

$$\text{s}^{-1} = \frac{1}{\text{s}}$$

means “per second.”

Similarly,

$$\text{h}^{-1} = \frac{1}{\text{h}}$$

means “per hour.”

3 Physical Half-Life

Physical half-life is the time required for one-half of the radioactive nuclei in a sample to decay.

It is represented by

$$T_{1/2}.$$

The relationship between physical half-life and the decay constant is

$$T_{1/2} = \frac{\ln 2}{\lambda}$$

or

$$T_{1/2} = \frac{0.693}{\lambda}$$

3.1 Relationship with Decay Constant

Physical half-life and decay constant are inversely related.

$$\boxed{T_{1/2} \uparrow \implies \lambda \downarrow}$$

and

$$\boxed{T_{1/2} \downarrow \implies \lambda \uparrow}$$

A radionuclide with a **short physical half-life** has a relatively large decay constant.

A radionuclide with a **long physical half-life** has a relatively small decay constant.

4 Technetium-99m Example

Technetium-99m has a physical half-life of approximately

$$T_{1/2} = 6 \text{ hours.}$$

Suppose a Tc-99m sample contains

$$N = 1.0 \times 10^{15}$$

radioactive nuclei.

We want to determine the activity of the sample.

4.1 Step 1: Convert Hours to Seconds

There are

$$3600 \text{ s}$$

in one hour.

Therefore,

$$6 \text{ h} \left(\frac{3600 \text{ s}}{1 \text{ h}} \right) = 21600 \text{ s.}$$

Thus,

$$\boxed{T_{1/2} = 21600 \text{ s}}$$

4.2 Step 2: Calculate the Decay Constant

Use

$$\lambda = \frac{\ln 2}{T_{1/2}}.$$

Substitute the half-life:

$$\lambda = \frac{0.693}{21600 \text{ s}}.$$

Therefore,

$$\boxed{\lambda = 3.21 \times 10^{-5} \text{ s}^{-1}}$$

4.3 Step 3: Calculate Activity

Use

$$A = \lambda N.$$

Substitute the values:

$$A = (3.21 \times 10^{-5} \text{ s}^{-1}) (1.0 \times 10^{15}).$$

When multiplying powers of ten,

$$10^{-5} \times 10^{15} = 10^{10}.$$

Therefore,

$$\boxed{A = 3.21 \times 10^{10} \text{ Bq}}$$

This is

$$32,100,000,000 \text{ Bq}.$$

In other words, approximately

$$\boxed{32.1 \text{ billion disintegrations occur every second}}$$

in the sample.

5 Becquerel

The Becquerel, abbreviated as

Bq,

is the SI unit of radioactive activity.

By definition,

$$1 \text{ Bq} = 1 \text{ disintegration per second}$$

or

$$1 \text{ Bq} = 1 \text{ s}^{-1}$$

when referring to radioactive transformations.

Therefore,

$$3.21 \times 10^{10} \text{ Bq}$$

means approximately

$$3.21 \times 10^{10}$$

radioactive disintegrations occur each second.

6 Disintegrations per Minute

DPM stands for

$$\text{Disintegrations Per Minute}$$

It expresses radioactive activity in terms of the number of nuclear disintegrations occurring every minute.

Since

$$1 \text{ minute} = 60 \text{ seconds},$$

and

$$1 \text{ Bq} = 1 \text{ disintegration per second},$$

then

$$1 \text{ Bq} = 60 \text{ dpm}$$

For example,

$$100 \text{ Bq} \times 60 = 6000 \text{ dpm}.$$

Therefore,

$$100 \text{ Bq} = 6000 \text{ dpm}$$

7 Curie and Radioactivity

7.1 Marie Curie

Marie Curie is associated with the discovery of the radioactive elements **radium** and **polonium**.

She also coined the term

radioactivity

to describe the phenomenon of spontaneous radiation emission from unstable atoms.

7.2 Radium and Polonium

Two important elements associated with the work of Marie Curie are

- **Radium**
- **Polonium**

These discoveries became important to the early understanding of radioactivity.

7.3 The Curie

The Curie is a unit of radioactive activity and is abbreviated as

Ci.

One Curie corresponds to

$$1 \text{ Ci} = 3.70 \times 10^{10} \text{ disintegrations per second}$$

Since

$$1 \text{ Bq} = 1 \text{ disintegration per second},$$

we can write

$$1 \text{ Ci} = 3.70 \times 10^{10} \text{ Bq}$$

The Curie was historically based on the activity associated with approximately

1 g

of radium-226.

8 Amount Remaining After Multiple Half-Lives

After every physical half-life, one-half of the radioactive nuclei remain.

The amount remaining can be calculated using

$$N = \frac{N_0}{2^n}$$

where

- N_0 = original number of radioactive nuclei
- N = number of radioactive nuclei remaining
- n = number of half-lives that have passed

The number of half-lives can be calculated using

$$n = \frac{t}{T_{1/2}}$$

8.1 Example

Suppose

$$N_0 = 800,$$

the physical half-life is

$$T_{1/2} = 6 \text{ hours},$$

and the total elapsed time is

$$t = 18 \text{ hours}.$$

First determine the number of half-lives:

$$n = \frac{18}{6} = 3.$$

Therefore,

$$N = \frac{800}{2^3}.$$

Since

$$2^3 = 8,$$

we obtain

$$N = \frac{800}{8} = \boxed{100}.$$

This can also be visualized as

$$800 \xrightarrow{6\text{ h}} 400 \xrightarrow{6\text{ h}} 200 \xrightarrow{6\text{ h}} 100.$$

The decay does not stop at 100.

If another half-life passes,

$$100 \rightarrow 50,$$

and another gives

$$50 \rightarrow 25.$$

9 Key Formulas

9.1 Activity

$$A = -\frac{dN}{dt}$$

$$A = \lambda N$$

9.2 Decay Constant

$$\lambda = \frac{\ln 2}{T_{1/2}} = \frac{0.693}{T_{1/2}}$$

9.3 Physical Half-Life

$$T_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}$$

9.4 Number Remaining

$$N = \frac{N_0}{2^n}$$

with

$$n = \frac{t}{T_{1/2}}$$

9.5 Activity Units

$$1 \text{ Bq} = 1 \text{ disintegration per second}$$

$$1 \text{ Bq} = 60 \text{ dpm}$$

$$1 \text{ Ci} = 3.70 \times 10^{10} \text{ Bq}$$

10 Radioactive Decay: Foundations and Classification

This addition is a detailed discussion guide for nuclear medicine. Learn the core clinical modes first: alpha, beta minus, beta plus, electron capture, and nuclear de-excitation by gamma emission or internal conversion. Later sections cover rare particle-emission and double-weak processes. Some names describe a primary transformation, others a subsequent step; they are not all mutually exclusive categories.

10.1 How to read a nuclear equation

Write a nuclide as A_ZX , where Z is proton number, A is nucleon (mass) number, and neutron number is $N = A - Z$. Here A means *mass number*; in the earlier activity formulas, A means *activity*. Context matters.

An asterisk denotes an excited nuclear state; m denotes a metastable state that persists long enough to be treated as an isomer. Isotopes share Z , isobars share A , isotones share N , and nuclear isomers share A and Z but differ in nuclear energy state.

For every equation, check electric charge and nucleon number on both sides. Also conserve total energy, momentum, and angular momentum. An emitted electron has charge -1 but mass number zero: mass number counts nucleons, not all particles with mass. Neutrinos have mass number and electric charge zero.

10.2 Why decay happens and what the Q-value means

An unstable parent can transform to a lower-total-energy configuration. Excess neutrons often favor beta minus decay; proton excess often favors beta plus or electron capture. Very heavy nuclei may favor alpha emission or fission. An excited nucleus can lose excitation without changing its element. These are guides, not substitutes for actual masses and nuclear selection rules.

The energy release is $Q = (m_{\text{initial}} - m_{\text{final}})c^2$, using a consistent accounting of all participating masses. A spontaneous channel requires positive available energy, but a positive Q alone does not guarantee a fast transition. Barriers, nuclear structure, and selection rules strongly influence its probability.

The daughter may be radioactive, stable, or excited. Disappearance of the parent does not mean all radioactivity has vanished. Different possible daughter states give different branches with their own energies and probabilities. Use evaluated decay data when exact numbers matter.

10.3 Nuclear-change reference

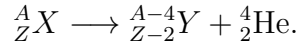
Process	Daughter A	Daughter Z	Daughter N
Alpha	$A - 4$	$Z - 2$	$N - 2$
Beta minus	A	$Z + 1$	$N - 1$
Beta plus / electron capture	A	$Z - 1$	$N + 1$
Gamma / internal conversion	A	Z	N
Internal pair creation	A	Z	N
One neutron emission	$A - 1$	Z	$N - 1$
One proton emission	$A - 1$	$Z - 1$	N
Two-neutron emission	$A - 2$	Z	$N - 2$
Two-proton emission	$A - 2$	$Z - 2$	N
Double beta minus	A	$Z + 2$	$N - 2$
Double EC / $EC\beta^+$ / $2\beta^+$	A	$Z - 2$	$N + 2$
Cluster ${}_Z^aC$	$A - a$	$Z - z$	$N - a + z$
Spontaneous fission	Partition among fragments and particles		

Gamma, conversion, and pair creation describe nuclear de-excitation. They do not change the chemical element.

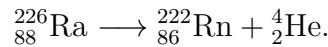
11 Alpha Decay

11.1 Mechanism, equation, and example

An alpha particle is a helium-4 nucleus: two protons plus two neutrons, with charge $+2e$. It initially has no bound electrons.



For example,



The daughter is a different element. Alpha emission is characteristic of many heavy radionuclides, though it is not confined to a single strict mass-number boundary. Escape through the Coulomb barrier is explained by quantum tunneling.

11.2 Energy spectrum and recoil

For a particular transition to a specified daughter state, the alpha energy is discrete. Several daughter levels can produce several alpha lines. Energy is shared between the alpha and recoiling daughter:

$$Q = T_\alpha + T_Y, \quad T_\alpha \simeq Q \frac{M_Y}{M_Y + m_\alpha}.$$

The heavy daughter receives less kinetic energy but is not motionless. If the daughter is excited, gamma emission or conversion can follow.

11.3 Nuclear medicine significance

Alpha particles have high linear energy transfer (LET): they deposit substantial energy over a short track, typically tens of micrometres in soft tissue for therapeutic alpha energies. Range depends on energy and material, so it is not one universal value. This supports targeted alpha therapy; radium-223 is a clinical example and actinium-225 is important in targeted-therapy research and development.

Low penetration reduces external exposure through intact skin, but internally deposited alpha emitters can cause intense local irradiation. Daughter recoil can disrupt a radionuclide's chemical attachment; the location and decay of radioactive daughters therefore matter.

Meeting answer. Alpha decay ejects a helium nucleus, so mass number falls by four and atomic number by two. It gives discrete alpha energies for each branch, high LET, and a short tissue range.

12 Beta Minus Decay

12.1 Mechanism and conservation

At the nucleon level,

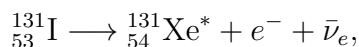
$$n \longrightarrow p + e^- + \bar{\nu}_e.$$

At the nuclide level,

$${}_Z^AX \longrightarrow {}_{Z+1}^AY + e^- + \bar{\nu}_e.$$

One neutron becomes a proton through the weak interaction. A is unchanged, Z increases by one, and N decreases by one. The electron is created in the transformation; it is not an orbital electron that was stored inside the nucleus. An electron antineutrino is emitted.

An important example is



with subsequent daughter de-excitation on appropriate branches. Other nuclear medicine beta emitters include yttrium-90 and lutetium-177.

12.2 Why the spectrum is continuous

The decay energy is shared among the electron, antineutrino, and recoil. Consequently the electron has a continuous distribution up to a branch-specific endpoint, rather than the same energy in every event. The endpoint is not the mean energy. Different populated daughter levels give different endpoints.

Using neutral atomic masses and neglecting small atomic corrections,

$$Q_{\beta^-} = [M_{\text{atom}}(X) - M_{\text{atom}}(Y)]c^2.$$

Subtract daughter excitation energy when considering an excited-state branch. The antineutrino usually escapes without depositing energy in the patient or detector.

12.3 Allowed and forbidden transitions

These labels classify angular-momentum and parity requirements; they do not change the beta-minus, beta-plus, or EC rules for A and Z . For allowed beta transitions, the emitted leptons carry zero orbital angular momentum. Allowed Fermi transitions have $\Delta J = 0$ with no parity change. Allowed Gamow–Teller transitions have $\Delta J = 0, 1$ with no parity change, excluding $0 \rightarrow 0$. Higher-order contributions permit forbidden transitions, often at reduced rates. The word forbidden therefore does not mean impossible.

Transition strength and available energy both matter. An allowed transition is not guaranteed to have a short half-life, and a higher energy alone does not determine the decay rate. Evaluated $\log ft$ values help characterize beta-transition strength.

12.4 Clinical and radiation implications

Beta particles deposit energy through ionization and excitation and usually travel farther than therapeutic alpha particles. Their finite range creates a cross-fire effect: energy can reach neighboring cells beyond the cell that contains the radionuclide. Its extent depends on the spectrum, tissue, and source distribution.

Electrons can also emit bremsstrahlung when decelerated in matter, especially in high- Z material. Bremsstrahlung is an interaction after emission, not a separate nuclear decay mode. Low- Z shielding is often used nearest a strong beta source, with additional shielding selected for any resulting photons and other emissions.

Meeting answer. Beta minus converts a neutron into a proton and emits an electron and an antineutrino. Mass number stays fixed, atomic number increases, and the electron spectrum is continuous because energy is shared.

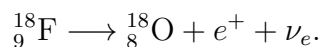
13 Beta Plus Decay and Positron Annihilation

13.1 Primary nuclear transformation

$$p \longrightarrow n + e^+ + \nu_e, \quad {}^A_Z X \longrightarrow {}^A_{Z-1} Y + e^+ + \nu_e.$$

Inside an energetically suitable nucleus, a proton converts to a neutron. A stays fixed, Z decreases by one, and N increases by one. A positron and an electron neutrino are emitted. This equation does not imply that an isolated free proton can spontaneously undergo this decay; the nuclear mass balance must permit it.

For example,



Fluorine-18 has a half-life of about 110 minutes and also has an electron-capture branch. Carbon-11, nitrogen-13, oxygen-15, gallium-68, and rubidium-82 are other useful PET examples.

13.2 The 1.022 MeV threshold

With neutral atomic masses,

$$Q_{\beta^+} = [M_{\text{atom}}(X) - M_{\text{atom}}(Y) - 2m_e]c^2.$$

Thus the parent-daughter atomic mass-energy difference must exceed $2m_e c^2 = 1.022$ MeV for a ground-state positron-emission branch. Excited-state branches require additional energy. The factor of two arises from atomic-electron bookkeeping as well as creating the positron; it does not mean the nucleus emits two positrons. The available kinetic energy is shared, so the positron spectrum is continuous.

13.3 What the PET scanner detects

The positron slows through interactions and then usually annihilates with an electron. For approximately stationary two-photon annihilation,

$$e^+ + e^- \longrightarrow \gamma + \gamma, \quad E_\gamma \simeq 511 \text{ keV}$$

for *each* photon. The photons travel in nearly opposite directions to conserve momentum. PET uses coincident detection to define a line of response. It does not directly image the emitted positron.

The 511 keV photons are produced by annihilation, not by a 511 keV nuclear level transition in fluorine-18. Positron travel before annihilation and slight non-collinearity contribute to resolution limits. Annihilation in flight and three-photon processes also exist, so exactly two back-to-back photons is the useful dominant model, not a description of every possible event.

Meeting answer. Beta plus decreases atomic number by one and emits a positron and a neutrino. The positron later annihilates, usually producing two approximately opposed 511 keV photons for PET.

14 Electron Capture

14.1 Mechanism and example

A proton interacts with an atomic electron whose wavefunction overlaps the nucleus:

$$p + e^- \longrightarrow n + \nu_e, \quad {}^A_Z X + e^- \longrightarrow {}^A_{Z-1} Y + \nu_e.$$

A stays fixed, Z decreases by one, and N increases by one. Capture commonly occurs from the K shell, but L and other shells can contribute. It is not an electron physically spiraling into the nucleus.

For example,

$${}^{123}_{53}\text{I} + e^- \longrightarrow {}^{123}_{52}\text{Te}^* + \nu_e.$$

Iodine-123 is useful for imaging because subsequent nuclear and atomic relaxation produces detectable photons. Indium-111 and thallium-201 are other familiar electron-capture radionuclides.

14.2 Energetics and competition with positron emission

The total atomic mass-energy release is approximately

$$Q_{\text{EC}} = [M_{\text{atom}}(X) - M_{\text{atom}}(Y)]c^2.$$

For a particular daughter level and captured shell, neutrino energy is approximately $Q_{\text{EC}} - E^* - B_i - E_{\text{recoil}}$, where B_i accounts for the atomic vacancy energy. Capture from a specified shell and to a specified state therefore gives an approximately discrete neutrino energy.

Electron capture does not have the 1.022 MeV positron-production threshold. It can occur when beta plus is energetically forbidden, provided its own final state is accessible. When both modes are possible, they can compete; the branching ratio is nuclide dependent.

14.3 The atomic vacancy

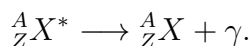
Capture leaves a vacancy in an electron shell. An outer electron fills it, releasing energy as a characteristic X-ray or transferring energy to another electron, which escapes as an Auger electron. Further vacancies can produce a cascade. A daughter nuclear excited state may separately emit gamma rays or conversion electrons.

Distinguish the two relaxation systems: X-rays and Auger electrons come from *atomic-shell* rearrangement; gamma rays come from *nuclear* level transitions. Electron capture emits no primary beta particle. It can nevertheless lead to emitted electrons through these secondary processes.

Meeting answer. Electron capture uses an orbital electron to convert a proton into a neutron. It lowers atomic number by one, emits a neutrino, and leaves an atomic vacancy that can produce characteristic X-rays or Auger electrons.

15 Gamma Emission and Isomeric Transition

15.1 Energy changes without transmutation



A gamma photon carries away nuclear excitation energy. Proton and neutron numbers do not change. Photon energy is approximately the level spacing, with a small recoil correction. Individual transitions give discrete photon energies; cascades give multiple lines.

An excited daughter created by alpha or beta decay may de-excite almost immediately. A metastable state has a delayed transition because its decay is hindered by its nuclear structure. The name *isomeric transition* (IT) describes de-excitation of an isomer, which can proceed through gamma emission and/or internal conversion. IT is not synonymous with gamma emission in every event.

15.2 Technetium-99m and SPECT

The simplified transformation is



Tc-99m has an approximately six-hour half-life and a prominent approximately 140.5 keV photon useful for gamma-camera/SPECT imaging. Its actual level scheme includes a cascade and conversion; not every parent decay gives one 140.5 keV photon. Tc-99 is still radioactive, with a much longer half-life, so a ground state need not be stable.

SPECT detects emitted single photons through a collimator, whereas PET uses coincidence detection of annihilation photons. Both rely on photons escaping the body. Gamma photons have zero charge and interact indirectly through secondary charged particles; they are attenuated probabilistically rather than having a sharp finite range.

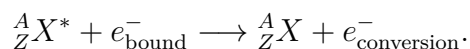
15.3 Gamma rays versus X-rays

The defining distinction is origin, not simply energy. Gamma rays come from nuclear transitions; characteristic X-rays come from electronic transitions. Their energy ranges can overlap. Identical-energy photons interact with matter in the same way regardless of that label.

16 Internal Conversion and Internal Pair Creation

16.1 Internal conversion: an alternative to a gamma photon

An excited nucleus transfers energy directly to a bound atomic electron, which is ejected:



This is schematic electron bookkeeping, not capture of an extra incoming electron. A and Z remain unchanged. A real gamma photon is not first emitted and then absorbed; conversion and photon emission are alternative channels for the nuclear transition.

For conversion from shell i ,

$$T_{e,i} \simeq E_{\text{transition}} - B_i.$$

The transition must supply enough energy to remove that electron. The resulting electron energies are discrete for a given nuclear transition and shell. Vacancies subsequently yield X-rays and Auger electrons, as in electron capture.

The internal conversion coefficient for a transition is

$$\alpha_{\text{IC}} = \frac{\text{conversion-electron transitions}}{\text{gamma-photon transitions}}.$$

If those are the only two channels, $P_{\text{IC}} = \alpha_{\text{IC}}/(1 + \alpha_{\text{IC}})$ and $P_{\gamma} = 1/(1 + \alpha_{\text{IC}})$. For example, a hypothetical $\alpha_{\text{IC}} = 0.25$ gives 20% conversion and 80% gamma emission. This is a probability for that transition, not automatically an emission yield per parent decay.

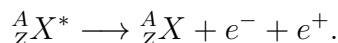
16.2 Electron capture versus conversion versus beta minus

- **Electron capture:** an existing electron is consumed, a proton becomes a neutron, Z decreases, and a neutrino is emitted.
- **Internal conversion:** an existing electron is ejected, the nucleus loses excitation, and Z does not change.
- **Beta minus:** an electron is created with an antineutrino, a neutron becomes a proton, and Z increases.

Auger electrons are a fourth category: they are ejected by *atomic* relaxation. Knowing only that an electron was emitted is insufficient to diagnose beta minus decay.

16.3 Internal pair creation

With sufficient nuclear transition energy, de-excitation can instead create an electron-positron pair:



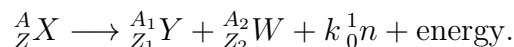
A and Z do not change. The transition energy must exceed approximately 1.022 MeV, and the excess becomes kinetic and recoil energy. The two leptons share the kinetic energy; their individual energies need not be discrete. The positron can subsequently annihilate.

This is not beta plus: there is no proton-to-neutron conversion and no required neutrino. It is also distinct from *external pair production*, where an already-existing high-energy photon interacts with matter. Electric-monopole ($E0$), $0^+ \rightarrow 0^+$ nuclear transitions cannot emit a single gamma photon; conversion and, when energetically allowed, internal pair creation are important alternatives. Rare two-photon nuclear de-excitation is another possible electromagnetic channel, with two photons sharing one transition's energy.

17 Spontaneous Fission and Cluster Radioactivity

17.1 Spontaneous fission

A heavy nucleus can split without an incoming projectile:

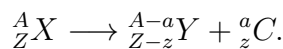


For this simplified binary channel, $A = A_1 + A_2 + k$ and $Z = Z_1 + Z_2$. There is a distribution of fragment masses, kinetic energies, and neutron multiplicities, not one unique daughter or a single fixed neutron count. Excited fragments emit prompt radiation; radioactive products may later undergo beta decay and produce delayed radiation.

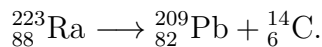
Californium-252 is a familiar spontaneous-fission neutron source, although alpha decay is its dominant decay branch. Ternary fission also occurs, with an additional light charged fragment. Spontaneous fission is a decay mode; neutron-induced fission is a nuclear reaction initiated by neutron absorption. Reactor fission can produce medically useful nuclides such as molybdenum-99, but that production route must not be mislabeled spontaneous decay.

17.2 Cluster emission

A heavy nucleus can emit a bound nuclear cluster heavier than an alpha:



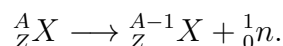
A classic example is



Charge and mass number balance: $88 = 82 + 6$ and $223 = 209 + 14$. Cluster radioactivity is a strongly asymmetric fragmentation channel; distinguish it from both alpha decay and ordinary fission. It is rare and has no routine role as a clinical imaging mode.

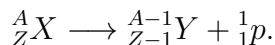
18 Nucleon Emission and Delayed Particle Decay

18.1 Neutron emission



The element stays the same, but A and N fall by one. This can occur from neutron-unbound states or very neutron-rich systems. A neutron has no Coulomb barrier; energetic accessibility and nuclear structure determine the emission. Most ordinary neutron-rich radionuclides should not simply be assumed to emit neutrons: many undergo beta minus. Free-neutron beta decay is a separate weak process.

18.2 Proton emission and multiple-nucleon emission



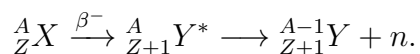
A and Z fall by one, while N stays fixed. Proton-rich unbound nuclei can emit a proton by penetrating the Coulomb barrier. Nuclear proton emission does not mean hypothetical decay of the proton itself.

Two-proton emission gives $(A-2, Z-2)$; two-neutron emission gives $(A-2, Z)$. Emission can be correlated/direct or proceed sequentially through intermediate states, depending on the accessible energies. Higher-multiplicity nucleon emission is also possible in very unbound systems. These are specialist nuclear-structure processes, not ordinary PET or SPECT decay routes.

18.3 Beta-delayed particles: two steps

Beta decay can populate a daughter above a particle-separation threshold. That daughter then emits a neutron, proton, alpha, multiple particles, or, in heavy systems, undergoes fission.

For beta-delayed neutron emission,



The first arrow also emits $e^- + \bar{\nu}_e$. A classic example is Br-87 beta decay feeding states in Kr-87 that can emit a neutron to Kr-86. Some branches may instead remain particle bound.

For a beta-plus-delayed proton branch, the final residual nucleus has $(A-1, Z-2)$ relative to the initial parent: the first step lowers Z by one and the second removes a proton. Electron capture can also feed particle-unbound states. The delay generally reflects the parent's weak-decay waiting time; emission from the unbound daughter is often prompt. Delayed neutrons are particularly important in reactor kinetics.

Rare beta-delayed deuteron ($d = {}^2\text{H}$) and triton ($t = {}^3\text{H}$) emission are additional light-particle channels. For example, He-6 can undergo a rare beta-delayed deuteron breakup into an alpha particle and a deuteron, with an electron and antineutrino from the weak transformation. Balance the beta step and the breakup separately. Particle-unbound light nuclei can also break up into alpha particles; Be-8 decaying into two alphas is a familiar example. These processes should not be confused with emitting a single free neutron or proton.

19 Double-Weak Decay and Other Rare Variants

19.1 Two-neutrino double beta minus

For some nuclei, single beta decay is energetically blocked or strongly suppressed while a second-order weak transition to $(A, Z+2)$ is possible:

$${}^A_Z X \longrightarrow {}^A_{Z+2} Y + 2e^- + 2\bar{\nu}_e.$$

This observed process leaves A fixed, increases Z by two, and decreases N by two. Examples include Ge-76 to Se-76 and Xe-136 to Ba-136. The electrons share energy with antineutrinos, so their summed kinetic energy is continuous. It is not simply two ordinary, temporally separate beta decays through a populated intermediate daughter.

19.2 Proton-to-neutron double-weak channels

These change Z by -2 at fixed A :

$$\begin{aligned} 2\beta^+ : \quad & {}^A_Z X \rightarrow {}^A_{Z-2} Y + 2e^+ + 2\nu_e, \\ \text{EC}\beta^+ : \quad & {}^A_Z X + e^- \rightarrow {}^A_{Z-2} Y + e^+ + 2\nu_e, \\ 2\text{EC} : \quad & {}^A_Z X + 2e^- \rightarrow {}^A_{Z-2} Y + 2\nu_e. \end{aligned}$$

For neutral atomic mass differences, double positron emission requires more than $4m_e c^2 = 2.044$ MeV; EC plus positron emission requires more than approximately 1.022 MeV, with shell and excitation requirements also included. Double capture has no positron-creation threshold, but the final nuclear and atomic state must still be energetically accessible. Double electron capture has been observed, for example in Xe-124. Do not infer that every conceivable double-weak channel has been observed in every candidate nuclide. These processes have exceptionally long half-lives and are not routine medical imaging mechanisms.

19.3 Neutrinoless modes: keep hypotheses separate

The proposed neutrinoless double-beta-minus process is

$${}^A_ZX \longrightarrow {}^A_{Z+2}Y + 2e^-.$$

It would violate lepton number; it must not be presented as an established decay observation. Unlike the two-neutrino mode, essentially all available energy would appear in the electrons, apart from recoil. Experiments search for a peak near the endpoint of their summed energy. Neutrinoless double capture and related positron channels are also research hypotheses. A recent LEGEND-200 report cited below found no signal; a non-observation constrains the half-life rather than proving the process impossible.

19.4 Bound-state beta decay and radiative variants

In bound-state beta minus decay, the newly created electron occupies a vacant bound orbital of the daughter rather than escaping as a free beta particle. The nuclear changes remain those of beta minus, and an antineutrino is emitted. This is especially relevant to highly ionized atoms; it is not the normal assumption for a radiopharmaceutical in tissue.

Rare radiative beta decay or radiative electron capture includes an additional photon in association with the weak process. Such photons must be distinguished from later daughter nuclear gamma rays and atomic characteristic X-rays. These are variants of the underlying weak mode, not additional changes to the A, Z rules.

20 Connecting Decay Modes to Activity and Imaging

20.1 Branching fractions and emitted radiation

If a parent has competing channels,

$$\lambda_{\text{total}} = \sum_i \lambda_i, \quad b_i = \frac{\lambda_i}{\lambda_{\text{total}}}, \quad A_i = b_i A_{\text{total}}.$$

There is one overall parent half-life: $T_{1/2} = \ln 2 / \lambda_{\text{total}}$. A branch's partial half-life is $T_{1/2,i} = T_{1/2} / b_i$, not a separate observed disappearance time for the same parent sample.

Activity counts transformations. Photon yield counts photons emitted per parent transformation. A cascade may emit more than one photon; conversion or other branches may suppress a particular line. For emission yield y and detection efficiency ϵ ,

$$\text{ideal net count rate} = A y \epsilon.$$

Efficiency here includes the relevant geometry and detection factors. Real measurements also require appropriate corrections for attenuation, scatter, background, and dead time.

Worked hypothetical example. A 20 MBq source has a photon yield of 0.80 photons per decay. Its photon emission rate is $20 \times 10^6 \times 0.80 = 16 \times 10^6$ photons/s. At a total detection efficiency of 0.001, the ideal count rate is 16,000 counts/s. Neither count rate nor photon rate equals activity automatically.

20.2 Decay chains and the Mo-99/Tc-99m generator

Mo-99 undergoes beta minus decay and feeds Tc-99m on a substantial branch. Tc-99m then undergoes isomeric de-excitation to Tc-99. In a generator, separation removes daughter activity; daughter production continues from the remaining parent.

For a two-member chain with parent feeding fraction b ,

$$\frac{dN_D}{dt} = b\lambda_P N_P - \lambda_D N_D.$$

For $N_D(0) = 0$ and unequal decay constants,

$$N_D(t) = \frac{b\lambda_P N_P(0)}{\lambda_D - \lambda_P} (e^{-\lambda_P t} - e^{-\lambda_D t}).$$

This explains why daughter activity can rise while parent activity falls. Secular equilibrium is approached when the parent half-life is much longer than the daughter's; transient equilibrium can occur when it is longer but not overwhelmingly so. A daughter longer lived than its parent does not establish these same equilibrium conditions.

20.3 Physical, biological, and effective half-life

The decay mode and physical half-life describe the radionuclide. Biological clearance describes removal from a tissue or body compartment. For independent, single-exponential decay and clearance,

$$\lambda_{\text{eff}} = \lambda_{\text{phys}} + \lambda_{\text{bio}}, \quad \frac{1}{T_{\text{eff}}} = \frac{1}{T_{\text{phys}}} + \frac{1}{T_{\text{bio}}}.$$

For hypothetical physical and biological half-lives of six and twelve hours, $T_{\text{eff}} = 4$ hours. Complex uptake and multi-compartment clearance need more detailed models. Biological removal does not change the radionuclide's nuclear identity.

20.4 Clinical comparison

- **SPECT:** useful escaping photons; examples include Tc-99m de-excitation and photons following I-123 electron capture.
- **PET:** positron emission followed by annihilation; F-18 and Ga-68 are familiar examples.
- **Beta therapy:** electron energy deposited over a finite range; examples include I-131, Y-90, and Lu-177.
- **Alpha therapy:** short tracks with high LET; Ra-223 is an example. Daughter emissions and redistribution must also be considered.
- **Auger-emitter strategies:** atomic electron cascades can deposit energy very locally. Biological effect depends strongly on subcellular localization, especially proximity to DNA.

These categories are not exclusive: a therapeutic radionuclide can also emit imaging photons. Clinical suitability depends on the entire decay scheme, half-life, chemical targeting, and biodistribution.

21 Processes That Are Often Mistaken for Decay Modes

- **Characteristic X-ray emission and Auger/Coster–Kronig emission:** atomic vacancy relaxation, often following EC or conversion. Coster–Kronig transitions involve subshells of the same principal shell.
- **Positron annihilation:** a particle-antiparticle interaction after positron production, not a change of nuclear A or Z .
- **Bremsstrahlung:** radiation from charged-particle deceleration in matter, not nuclear transmutation.
- **Photoelectric absorption, Compton scattering, and external pair production:** photon interactions in matter; distinguish them from the processes that produced the photons.
- **Neutron activation and induced fission:** reactions driven by an incoming particle; their products may subsequently decay.

22 Meeting Practice: Questions with Model Answers

1. **How do you identify the mode from a daughter nuclide?** Compare A and Z : $(-4, -2)$ suggests alpha, $(0, +1)$ beta minus, and $(0, -1)$ beta plus or EC. Distinguish the last two by emitted particles and energetics.
2. **Can a decay leave the element unchanged?** Yes. Gamma emission, conversion, and pair de-excitation leave both A and Z unchanged. Neutron emission leaves Z unchanged but reduces A .
3. **Why does beta decay need a neutrino or antineutrino?** The additional particle carries energy, momentum, and angular momentum; the full process also respects the lepton accounting of ordinary weak decay. It explains why beta energies are not all identical.
4. **Which beta process emits an antineutrino?** Beta minus emits $\bar{\nu}_e$; beta plus and EC emit ν_e .
5. **Why can EC occur when beta plus cannot?** EC does not need the 1.022 MeV neutral-atomic mass difference required for positron emission. Its selected final state must still be accessible.
6. **Does 511 keV equal the positron's initial kinetic energy?** No. It is each photon's energy in approximately at-rest two-photon annihilation, corresponding to the electron rest energy.

7. **Is internal conversion a beta decay?** No. Conversion ejects a pre-existing atomic electron and leaves nuclear Z unchanged; beta minus creates an electron and increases Z .
8. **Can electron capture produce electrons outside the atom?** Yes, indirectly: the vacancy can lead to Auger electrons, and an excited daughter can undergo conversion. No primary beta particle is emitted by the capture step.
9. **Are gamma rays always higher energy than X-rays?** No. Origin defines the names, and their energy ranges overlap.
10. **Does Tc-99m turn into another element?** Its usual IT route produces Tc-99, the same element and mass number in a lower nuclear state. Tc-99 itself remains radioactive.
11. **Does one Bq mean one detected photon per second?** No. It means one transformation per second; photon yield and detector efficiency determine the photon count rate.
12. **Why is an alpha source relevant despite short range?** When located within tissue it deposits energy densely near its location. Short external range does not imply negligible internal effect.
13. **What is the difference between endpoint and mean beta energy?** The endpoint is the maximum allowed electron kinetic energy for a branch; the mean is an average over its entire spectrum.
14. **Is a beta-delayed neutron emitted in the beta step?** No. Beta decay feeds an unbound daughter, which then emits a neutron.
15. **How does spontaneous fission differ from alpha decay?** Fission distributes the parent among large fragments, often plus neutrons. Alpha decay emits specifically a helium-4 nucleus.
16. **Is double beta decay the same as a decay chain?** No. Double beta is a second-order transition between nuclei differing by two charge units; an ordinary chain populates a real intermediate nuclide in successive decays.
17. **Has neutrinoless double beta decay been established?** No confirmed observation is established in the research sources cited here. Explain it as a searched-for process, not a clinical decay mode.
18. **What happens if two decay branches are available?** Their decay constants add. The total half-life reflects both routes; branching fractions describe the fraction following each one.
19. **Does heating normally change physical half-life?** Ordinary temperature and chemical changes generally have negligible effects for clinical calculations. EC and highly ionized atoms require care: electron availability can affect electron-dependent decay.

20. **How would you explain any decay in one minute?** State the driving instability, write the equation, give changes in A , Z , N , identify all primary and secondary radiations, describe the energy spectrum, and connect one example to nuclear medicine.

22.1 Quick equation drills

1. Po-210 emits an alpha: $A = 206$, $Z = 82$, so the daughter is Pb-206.
2. P-32 undergoes beta minus: $A = 32$, $Z = 16$, so the daughter is S-32, with an electron and antineutrino.
3. F-18 undergoes beta plus: $A = 18$, $Z = 8$, so the daughter is O-18, with a positron and neutrino.
4. I-123 captures an electron: $A = 123$, $Z = 52$, so the daughter is Te-123, often initially excited.
5. A hypothetical 200 keV transition converts a 30 keV bound electron: its kinetic energy is approximately 170 keV, neglecting recoil.
6. After three half-lives, 12.5% of the original parent remains. Its activity is also 12.5% if no parent is added or newly produced.

23 References and Further Study for the Decay Addition

The explanations and practice problems above are an original study synthesis. Numerical examples labeled hypothetical are teaching calculations, not measured nuclide data. Half-lives and energies quoted in the discussion are rounded. Consult the full evaluated decay scheme for precise branch intensities, uncertainties, and dosimetry.

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